Kameron Benjamin

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EDUCATION

Florida Agricultural and Mechanical University, B.S. in Computer Science, Major GPA: 3.3 *Aug 2023 – May 2027*
Relevant Coursework: Discrete Mathematics I & II, Object Oriented Programming, Data Networks & Communication, Database Management, Data Structures & Algorithms, Computer Ethics and Professional Responsibility, Algorithm and System Design, Intro to Software Engineering
Organizations: Google CodePath Tech Exchange, FAMU Marching "100" Band, Strikers Dance Troupe, Kappa Kappa Psi National Honorary Band Fraternity Inc., ColorStack

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks & Tools: FastAPI, Docker, AWS (EC2, Lambda, S3, CloudFront), GitHub Actions (CI/CD), Git, Linux, Proxmox, systemd, Node.js, React, Next.js, React Native, Three.js, REST APIs
AI / ML: PyTorch, Large Language Models (LLMs), Claude API, Agentic AI Pipelines, Prompt Engineering, Data Processing

WORK EXPERIENCES

UCSC Open Source Program Office, Contributor Catalyst **Santa Cruz, CA**
Open Source Intern | p5.js, Node.js, Git, GitHub, Open Source *May 2026 – Aug 2026*
- Researched packaging the p5.js reference as a downloadable offline release artifact, reviewing the site build and release workflow.
- Mentored fellow cohort participants on Git, GitHub, the command line, VS Code, and open source collaboration.
- Presented Green Guardian at the inaugural PyTorch Santa Cruz Meetup alongside NVIDIA and Red Hat speakers, and was featured in a PyTorch blog post.

Florida Agricultural and Mechanical University **Tallahassee, FL**
Teaching Assistant – Intro Object Oriented Programming in C++ | C++, Debugging, Data Structures *Jan 2026 – May 2026*
- Supported undergraduate OOP in C++ by reinforcing core concepts, improving student understanding of classes, inheritance, and polymorphism.
- Led three weekly problem-solving sessions for approximately 30 students, reducing debugging time by 10%.
- Assisted students in diagnosing runtime errors, segmentation faults, and logic issues while reinforcing OOP principles.

Outamation **Tallahassee, FL**
AI/Data Automation Extern | LLM, Prompt engineering, RAG Pipelines, Data Cleaning, PyMuPDF *Oct 2025 – Jan 2026*
- Built Python pipelines using LLMs to process and classify large volumes of unstructured business documents, cutting turnaround time by about 30%.
- Improved classification accuracy by about 25% by iterating on prompt engineering workflows that extract, classify, and summarize key information.
- Developed automated data-cleaning and parsing tools, iterating on solutions under ambiguous requirements to balance accuracy, performance, and usability.

PROJECTS

Homelab Agent Platform – Self-Hosted AI Coding Agents | github.com/kameron-ctrl/homelab-agent-platform *Oct 2026 – Present*
Python, Docker, Proxmox (LXC), systemd, FastAPI, Telegram Bot API, GitHub Actions
- Built a Telegram-driven harness that routes coding tasks to Claude, Codex, or Gemini CLI agents by learned success rate, failing over to another agent on rate limits.
- Isolated agent workloads in resource-limited Docker sandboxes on a firewalled network, with a 4-hour idle reaper that reclaims unused containers.
- Automated nightly dependency updates across 5 repositories, opening pull requests only after each repo's test gate passes.
- Secured access with forced-command SSH keys and allowlisted read-only exports that feed a FastAPI dashboard of agent jobs and token usage.

Green Guardian – Full Stack AI Plant Disease Detector | github.com/kameron-ctrl/greenguardian | greenguardian.dev *May 2025 – Oct 2025*
Python, FastAPI, PyTorch, React Native, Expo, Next.js, Docker, AWS (Lambda, ECR, CloudFront)
Web and mobile app that identifies 15 plant disease classes from leaf images in real time.
- Built an image classifier for 15 plant disease classes by fine-tuning a ResNet-18 on PlantVillage, serving real-time predictions via FastAPI.
- Designed RESTful APIs shared by a Next.js web app and a React Native/Expo mobile app, deployed on AWS Lambda, S3, and CloudFront.
- Cut deployment time by about 40% by automating builds and releases with GitHub Actions, including CloudFront cache invalidation.

CubeCoach – AI Rubik's Cube Trainer | github.com/kameron-ctrl/cubecoach *Mar 2026 – Present*
Next.js 14, TypeScript, Three.js, FastAPI, Python, Claude API, SQLite/PostgreSQL
- Built solo an AI Rubik's Cube trainer with an interactive Three.js cube and a FastAPI backend that uses Claude to explain each move.
- Turned any valid cube state into step-by-step teaching moves by wrapping a Kociemba solver and splitting each solution into steps.
- Built spaced-repetition drills for 78 algorithms (57 OLL + 21 PLL) using the SM-2 scheduler, with SQLAlchemy-backed progress tracking.

Personal Portfolio | kameron-ctrl.com *Jul 2026 – Present*
Next.js, TypeScript, Three.js, AWS (S3, CloudFront, ACM), GitHub Actions
- Migrated the site from a homelab server to S3 and CloudFront with ACM TLS, deployed by GitHub Actions through a keyless OIDC IAM role.

LEADERSHIP & CAMPUS INVOLVEMENT

Strikers Dance Troupe **Tallahassee, FL**
President, Assistant Rehearsal Director, Historian *Sept 2023 – Present*
- Led a 40+ member dance troupe as President, delivering 2 concerts per semester, by serving as the group's public face and main point of contact, managing rehearsal logistics and schedules, and coordinating officers so every event ran on time.
- Increased event reach and audience engagement by over 50% by managing and strategizing content across all social media platforms.

Kappa Kappa Psi National Honorary Band Fraternity Incorporated **Tallahassee, FL**
Vice President of Programs, Co-Webmaster *Apr 2025 – Present*
- Planned and executed chapter programming as Vice President of Programs by owning operational planning, coordinating service project execution, and keeping members and officers aligned through internal coordination.
- Maintained and updated the fraternity's website (deltaiota.org), improving accessibility and user experience.